SURVEY ON REAL TIME HELMET VIOLATION AND DETECTION USING DEEP LEARNING

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ABSTRACT

Artificial Intelligence (AI) has become a transformative technology for addressing real-world societal challenges by enabling machines to perform tasks that traditionally require human intelligence. In particular, deep learning has gained significant importance in the field of computer vision, where it supports automatic recognition, classification, and analysis of visual data. These capabilities have proven highly effective in traffic surveillance systems, enabling improved monitoring, decision-making, and enforcement to enhance road safety.

In existing research, an artificial intelligence–based multitier framework with a lightweight classifier was proposed for helmetless biker detection. The system follows a two-stage approach in which motorbike riders are first detected from traffic surveillance footage and then classified based on helmet usage. The use of a lightweight model helps reduce computational complexity and supports faster processing while maintaining acceptable accuracy. Custom datasets were also developed to evaluate the system under controlled conditions. However, the existing approach focuses primarily on helmet detection and does not address other critical traffic violations or challenging environmental scenarios.

To overcome these limitations, the proposed system employs DetectNet and the YOLOv12 algorithm to enable comprehensive and real-time traffic violation detection. In addition to helmet detection, the system is designed to identify multiple violations such as triple riding and mobile phone usage, while also maintaining robustness under diverse conditions including night-time and rainy environments. Furthermore, the system integrates an automated alert mechanism that notifies authorized institutions and sends warning messages to riders. This holistic approach improves detection accuracy, supports real-time enforcement, and enhances practical deployment, ultimately contributing to safer and more intelligent traffic management systems.

Keywords: Artificial Intelligence, Deep Learning, Helmet Detection, Traffic Safety, Computer Vision